

Name:

Student's ID:

114-1 Calculus (I) - Midterm Exam

Part I. MULTIPLE CHOICE (2 pts each)

For each problem, select exactly one answer. If you think there is no answer among the choices, then put an E.

E A 1. Find the function $f \circ g$ and its domain if $f(x) = \sqrt{x+6}$ and $g(x) = \sqrt{x-6}$.

(A) $\sqrt{x^2 - 36}$, $D = [6, \infty)$

(B) $\sqrt{x^2 - 36}$, $D = [-6, \infty)$

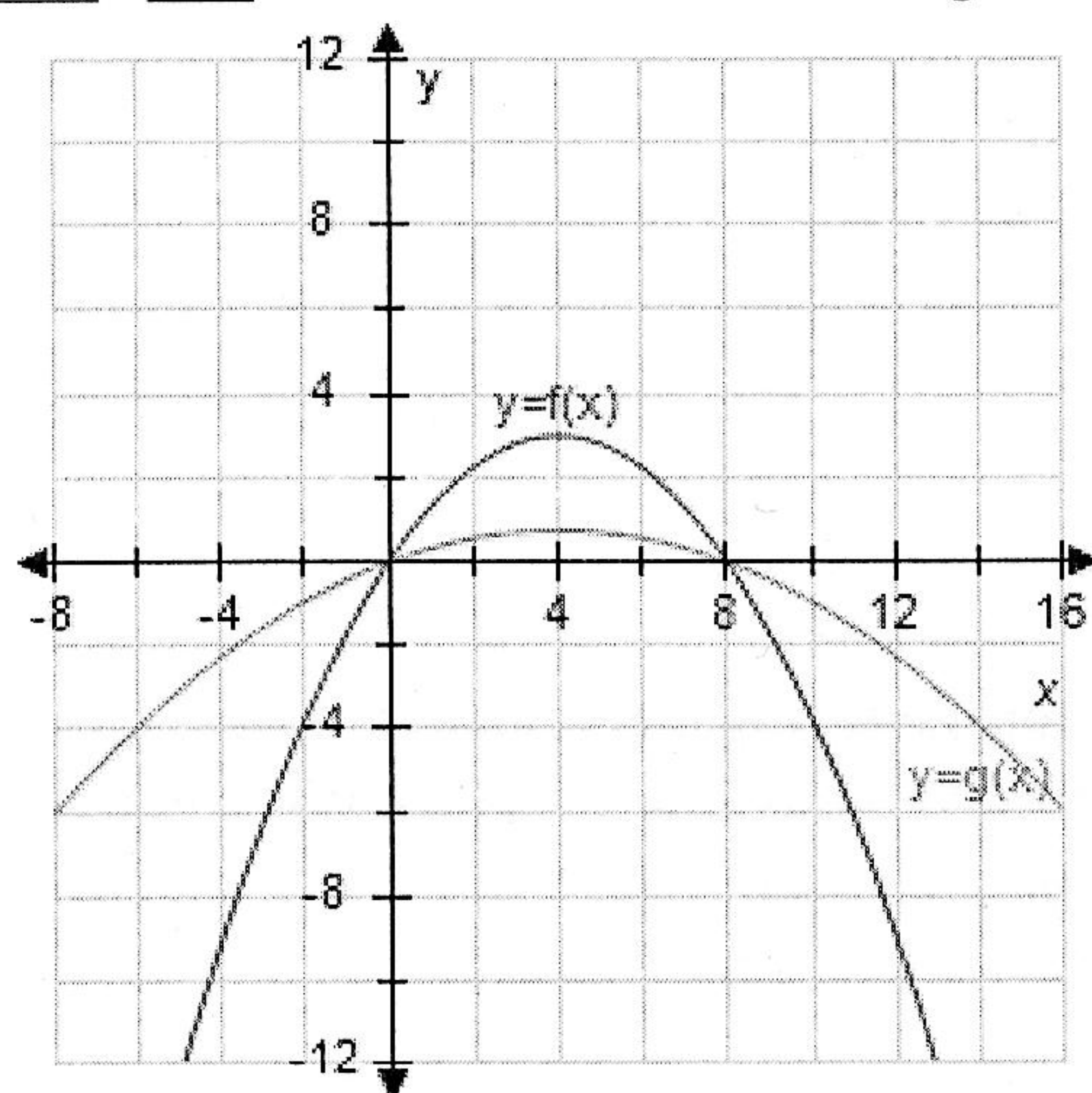
$(f \circ g)(x) = \sqrt{\sqrt{x-6} + 6}$

(C) $\sqrt{x^2 + 36}$, $D = [-6, \infty)$

(D) $\sqrt{x^2 + 36}$, $D = [6, \infty)$

$D = [6, \infty)$

D 2. Which of the following is the equation for the function $g(x)$?



(A) $g(x) = 4f(x)$

(B) $g(x) = f(x) - 4$

(C) $g(x) = -f(x+4)$

(D) $g(x) = f(x)/4$

D 3. 104 Let $F(x) = \frac{x-2}{|x-2|}$. Find the following limits. $\lim_{x \rightarrow 2^+} F(x)$, $\lim_{x \rightarrow 2^-} F(x)$.

(A) both 1

(B) 2 and 1

(C) 2 and -1

(D) 1 and -1

C 4. If a rock is thrown upward on the planet Mars with a velocity of 10 m/s, its height in meters t seconds later is given by $y = 10t - 1.9t^2$. Find the average velocity over the time interval $[2, 3]$.

(A) -2.5 m/s

(B) 2.5 m/s

(C) 0.5 m/s

(D) 1.5 m/s

B 5. The symbol $\lfloor \cdot \rfloor$ can be used to denote the greatest integer function, which is defined by $\lfloor x \rfloor$ the greatest integer n such that $n \leq x$. Use the graph of the function to find the indicated limit, if it exists.

$\lim_{x \rightarrow -1.5} \lfloor x \rfloor$

(A) -1

(B) -2

(C) 1.5

(D) -1.5

___B___ 6. Find equations of the tangent lines to the curve $y = \frac{x-7}{x+7}$ that are parallel to the line $x-y=7$. Round all numbers to one decimal place, if necessary.

(A) $x-y = -4.5$, $x-y = -15.5$ (B) $x-y = -0.5$, $x-y = -15.5$

(C) $x-y = -12.5$ (D) $x-y = -15.5$

___A___ 7. Compute Δy and dy for the given values of x and $dx=\Delta x$. $y = x^2$, $x = 5$, $\Delta x = 0.5$.

(A) $\Delta y = 5.25$, $dy = 5$ (B) $\Delta y = 6.05$, $dy = 0$

(C) $\Delta y = 5$, $dy = 0$ (D) $\Delta y = 5.25$, $dy = 6.05$

___A___ 8. Calculate y' . $xy^4 + x^3y = x + 4y$.

(A) $y' = \frac{1-y^4-3x^2y}{4xy^3+x^3-4}$ (B) $y' = \frac{1-y^4-2x^3}{4xy^2+x^2-4}$

(C) $y' = \frac{-y^4-3xy}{4xy^3+x^2}$ (D) $y' = \frac{xy^3+2x-4}{x^3y^2(4x-1)}$

___C___ 9. Suppose the daily total cost (in dollars) of manufacturing x televisions is $C(x) = 0.0005x^3 - 0.07x^2 + 160x + 6000$. What is the marginal cost when $x = 500$? What is the actual cost incurred in manufacturing the 501st television?

(A) \$465.00, \$465.93 (B) \$262.00, \$262.41

(C) \$465.00, \$465.68 (D) \$262.00, \$262.52

___B___ 10. An equation relating the variables x and y , the values of x and y , and the value of $\frac{dx}{dt}$ at a particular instant of time are given. Find the value of $\frac{dy}{dt}$.

$x^2y = 45$; $x = -3$, $y = 5$, $\frac{dx}{dt} = 4$; $\frac{dy}{dt} = ?$

(A) $\frac{45}{16}$

(B) $\frac{40}{3}$

(C) $-\frac{45}{16}$

(D) $-\frac{40}{3}$

Part II. Problems

For the following problems, write the details of the calculations or arguments on your answer sheets.

1.Ch1.5: E7

2.Ch1.6: E43

3.Ch1.7: E21 (quiz)

4.Ch1.8: E41

5.Ch2.1: E9

6.Ch2.2: E17

7.Ch2.4: E65

8.Ch2.5: E12, E47 (same with 113-1)

9.Ch2.9: E5

10.Ch3.1: E59 (quiz)

11.Ch3.2: E15

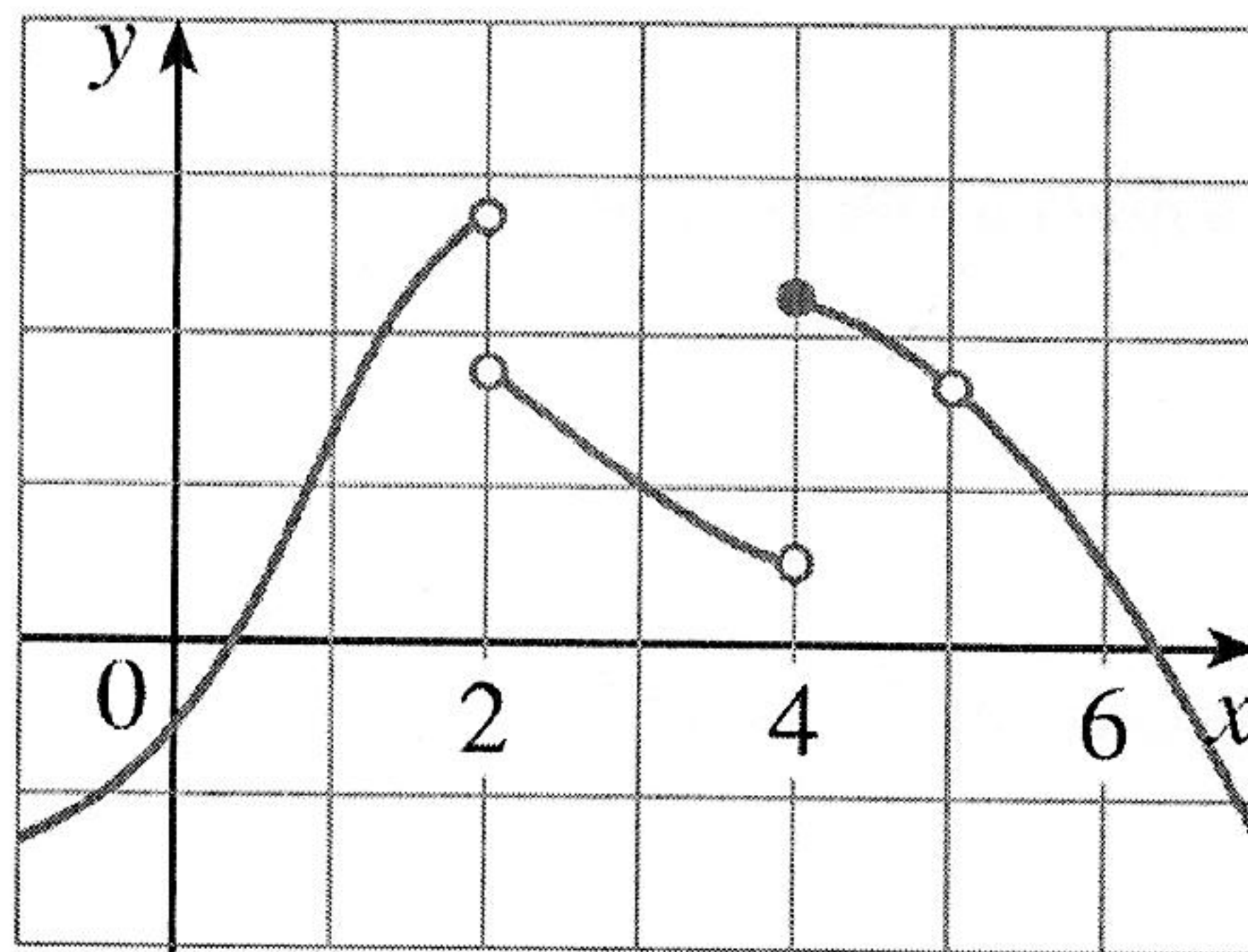
12.Ch3.3: E46

13.Ch3.4: E37,

14.Ch3.4: E55

1. (4 pts) For the function g whose graph is shown, find a number a that satisfies the given description.

- (a) $\lim_{x \rightarrow a} g(x)$ does not exist but $g(a)$ is defined.
- (b) $\lim_{x \rightarrow a} g(x)$ exists but $g(a)$ is not defined.
- (c) $\lim_{x \rightarrow a^-} g(x)$ and $\lim_{x \rightarrow a^+} g(x)$ both exist but $\lim_{x \rightarrow a} g(x)$ does not exist.
- (d) $\lim_{x \rightarrow a^+} g(x) = g(a)$ but $\lim_{x \rightarrow a^-} g(x) \neq g(a)$.



Answer:

- (a) $\lim_{x \rightarrow 4^-} g(x) \neq \lim_{x \rightarrow 4^+} g(x)$, so $\lim_{x \rightarrow 4} g(x)$ does not exist. However, there is a point on the graph representing $g(4)$. Thus, $a = 4$ satisfies the given description.
- (b) $\lim_{x \rightarrow 5^-} g(x) = \lim_{x \rightarrow 5^+} g(x)$, so $\lim_{x \rightarrow 5} g(x)$ exists. However, $g(5)$ is not defined. Thus, $a = 5$ satisfies the given description.
- (c) From part (a), $a = 4$ satisfies the given description. Also, $\lim_{x \rightarrow 2^-} g(x)$ and $\lim_{x \rightarrow 2^+} g(x)$ exist, but $\lim_{x \rightarrow 2^-} g(x) \neq \lim_{x \rightarrow 2^+} g(x)$. Thus, $\lim_{x \rightarrow 2} g(x)$ does not exist, and $a = 2$ also satisfies the given description. (2 or 4) 寫一個就對
- (d) $\lim_{x \rightarrow 4^+} g(x) = g(4)$, but $\lim_{x \rightarrow 4^-} g(x) \neq g(4)$. Thus, $a = 4$ satisfies the given description.

2. (4 pts) Find the limit, if it exists. If the limit does not exist, explain why.

$$\lim_{x \rightarrow -4} (|x + 4| - 2x)$$

Answer:

$$|x+4| = \begin{cases} x+4 & \text{if } x+4 \geq 0 \\ -(x+4) & \text{if } x+4 < 0 \end{cases} = \begin{cases} x+4 & \text{if } x \geq -4 \\ -(x+4) & \text{if } x < -4 \end{cases}$$

Thus, $\lim_{x \rightarrow -4^+} (|x+4| - 2x) = \lim_{x \rightarrow -4^+} (x+4 - 2x) = \lim_{x \rightarrow -4^+} (-x+4) = 4+4=8$ and

$$\lim_{x \rightarrow -4^-} (|x+4| - 2x) = \lim_{x \rightarrow -4^-} (-(x+4) - 2x) = \lim_{x \rightarrow -4^-} (-3x-4) = 12-4=8.$$

The left and right limits are equal, so $\lim_{x \rightarrow -4} (|x+4| - 2x) = 8$.

3. (4 pts) Prove the statement using the ε, δ definition of a limit.

$$\lim_{x \rightarrow 4} \frac{x^2 - 2x - 8}{x - 4} = 6$$

Answer:

Given $\varepsilon > 0$, we need $\delta > 0$ such that if $0 < |x - 4| < \delta$, then $\left| \frac{x^2 - 2x - 8}{x - 4} - 6 \right| < \varepsilon \Leftrightarrow$

$$\left| \frac{(x-4)(x+2)}{x-4} - 6 \right| < \varepsilon \Leftrightarrow |x+2-6| < \varepsilon \quad [x \neq 4] \Leftrightarrow |x-4| < \varepsilon. \text{ So choose } \delta = \varepsilon. \text{ Then}$$

$$0 < |x-4| < \delta \Rightarrow |x-4| < \varepsilon \Rightarrow |x+2-6| < \varepsilon \Rightarrow \left| \frac{(x-4)(x+2)}{x-4} - 6 \right| < \varepsilon \quad [x \neq 4] \Rightarrow$$

$$\left| \frac{x^2 - 2x - 8}{x - 4} - 6 \right| < \varepsilon. \text{ By the definition of a limit, } \lim_{x \rightarrow 4} \frac{x^2 - 2x - 8}{x - 4} = 6.$$

4. (6 pts) Show the f is continuous on $(-\infty, \infty)$.

$$f(x) = \begin{cases} 1 - x^2, & \text{if } x \leq 1 \\ \sqrt{x-1}, & \text{if } x > 1 \end{cases}$$

Answer:

$$f(x) = \begin{cases} 1 - x^2 & \text{if } x \leq 1 \\ \sqrt{x-1} & \text{if } x > 1 \end{cases}$$

* f is continuous on $(-\infty, 1) \cup (1, \infty)$ 絕對
寫 $f(x) = 1 - x^2$ is continuous on $(-\infty, 1)$ 扣 1

By Theorem 5, since $f(x)$ equals the polynomial $1 - x^2$ on $(-\infty, 1]$, f is continuous on $(-\infty, 1]$. By Theorem 7, since $f(x)$ equals the root function $\sqrt{x-1}$ on $(1, \infty)$, f is continuous on $(1, \infty)$. At $x = 1$, $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (1 - x^2) = 1 - 1^2 = 0$

and $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \sqrt{x-1} = \sqrt{1-1} = 0$. Thus, $\lim_{x \rightarrow 1} f(x)$ exists and equals 0. Also, $f(1) = 1 - 1^2 = 0$. Therefore, f is continuous at $x = 1$. We conclude that f is continuous on $(-\infty, \infty)$.

5. (4 pts)

(a) Find the slope of the tangent to the curve

$$y = 3 + 4x^2 - 2x^3 \text{ at the point where } x = a.$$

(b) Find the equation of the tangent lines at the point (2, 3).

Answer:

(a) Using (2) with $y = f(x) = 3 + 4x^2 - 2x^3$, the slope of the tangent line is

$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{3 + 4(a+h)^2 - 2(a+h)^3 - (3 + 4a^2 - 2a^3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3 + 4(a^2 + 2ah + h^2) - 2(a^3 + 3a^2h + 3ah^2 + h^3) - 3 - 4a^2 + 2a^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{3 + 4a^2 + 8ah + 4h^2 - 2a^3 - 6a^2h - 6ah^2 - 2h^3 - 3 - 4a^2 + 2a^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{8ah + 4h^2 - 6a^2h - 6ah^2 - 2h^3}{h} = \lim_{h \rightarrow 0} \frac{h(8a + 4h - 6a^2 - 6ah - 2h^2)}{h} \\ &= \lim_{h \rightarrow 0} (8a + 4h - 6a^2 - 6ah - 2h^2) = 8a - 6a^2 \end{aligned}$$

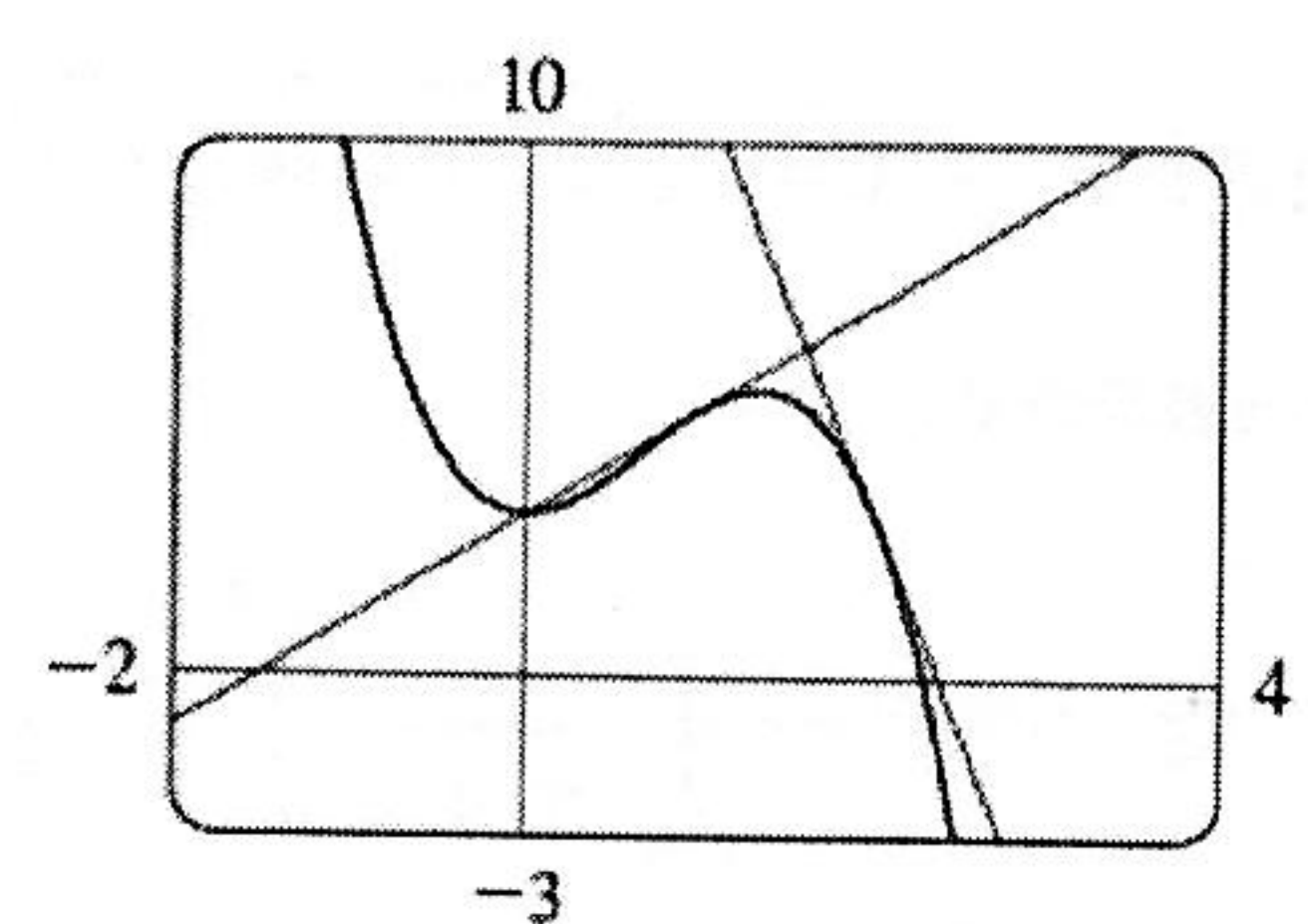
(b) At (1, 5): $m = 8(1) - 6(1)^2 = 2$, so an equation of the tangent line

$$\text{is } y - 5 = 2(x - 1) \Leftrightarrow y = 2x + 3.$$

At (2, 3): $m = 8(2) - 6(2)^2 = -8$, so an equation of the tangent

$$\text{line is } y - 3 = -8(x - 2) \Leftrightarrow y = -8x + 19.$$

(c)



6. (4 pts) Let $f(x) = x^2$.

1 (a) Estimate the values of $f'(0)$, $f'(\frac{1}{2})$, and $f'(2)$ by zooming in on the graph of f .

1 (b) Use symmetry to deduce the values of $f'(-\frac{1}{2})$, $f'(-1)$, and $f'(-2)$.

2 (c) Use the definition of derivative to prove the formula for $f'(x)$.

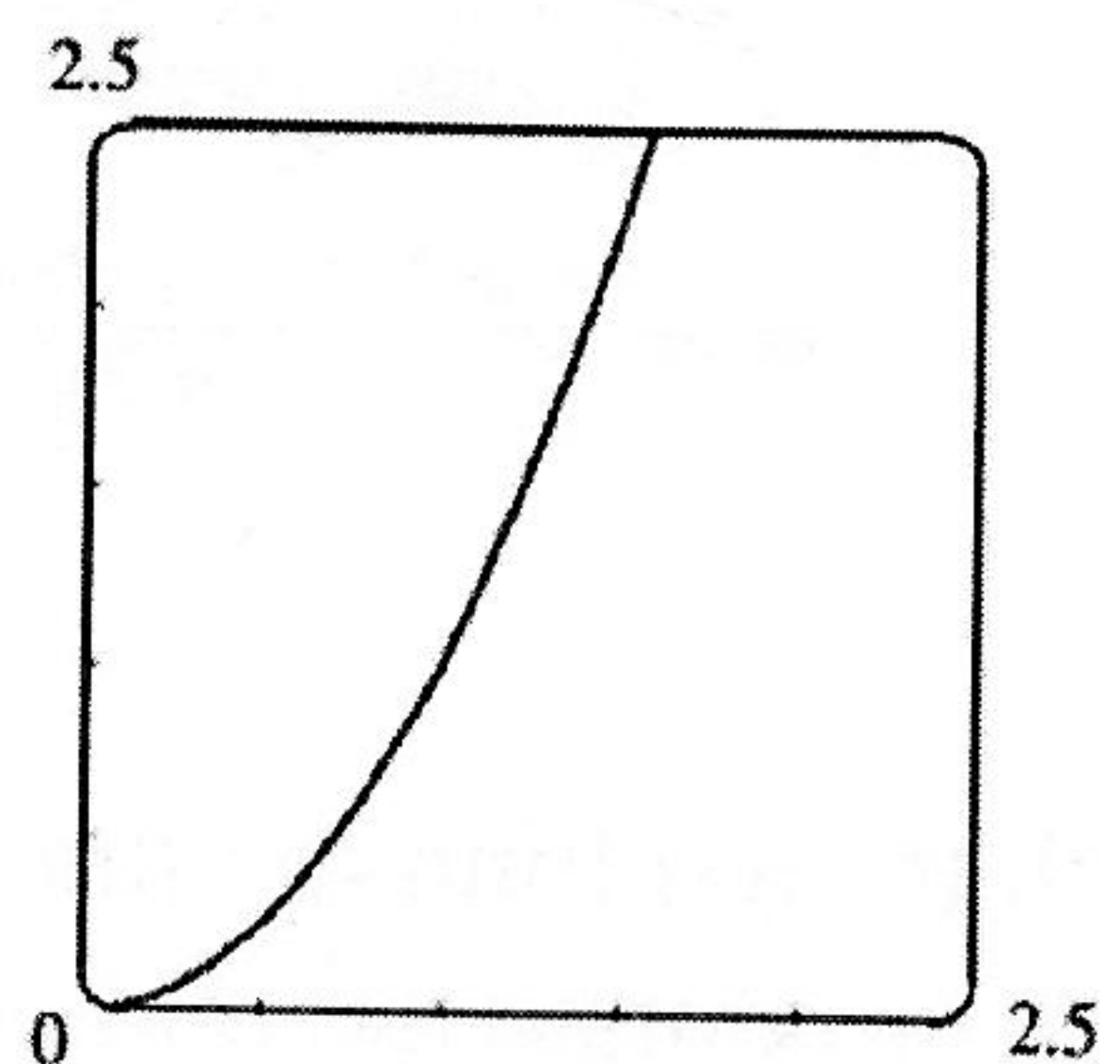
Answer:

(a) By zooming in, we estimate that $f'(0) = 0$, $f'(\frac{1}{2}) = 1$, $f'(1) = 2$,
and $f'(2) = 4$.

(b) By symmetry, $f'(-x) = -f'(x)$. So $f'(-\frac{1}{2}) = -1$, $f'(-1) = -2$,
and $f'(-2) = -4$.

(c) It appears that $f'(x)$ is twice the value of x , so we guess that $f'(x) = 2x$.

$$\begin{aligned} \text{(d) } f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x^2 + 2hx + h^2) - x^2}{h} = \lim_{h \rightarrow 0} \frac{2hx + h^2}{h} = \lim_{h \rightarrow 0} \frac{h(2x + h)}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x \end{aligned}$$



7. (6 pts) Differentiate the trigonometric identity to obtain a new (or familiar) identity.

$$\sin x + \cos x = \frac{1 + \cot x}{\csc x}$$

Answer:

$$(c) \frac{d}{dx} (\sin x + \cos x) = \frac{d}{dx} \frac{1 + \cot x}{\csc x} \Rightarrow$$

$$\begin{aligned} \cos x - \sin x &= \frac{\csc x (-\csc^2 x) - (1 + \cot x)(-\csc x \cot x)}{\csc^2 x} = \frac{\csc x [-\csc^2 x + (1 + \cot x) \cot x]}{\csc^2 x} \\ &= \frac{-\csc^2 x + \cot^2 x + \cot x}{\csc x} = \frac{-1 + \cot x}{\csc x} \end{aligned}$$

$$\text{So } \cos x - \sin x = \frac{\cot x - 1}{\csc x} \quad \text{邊程錯 -2} \quad * \quad \text{小地方寫錯只扣1 (±)}$$

8. (6 pts) Find the derivative of the function.

$$3 \text{ (a) } F(t) = \left(\frac{1}{2t+1}\right)^4. \quad \text{邊程錯 -2}$$

$$3 \text{ (b) } y = \cos \sqrt{\sin(\tan \pi x)} \quad \text{邊程錯 -2}$$

Answer:

$$12. F(t) = \left(\frac{1}{2t+1}\right)^4 = [(2t+1)^{-1}]^4 = (2t+1)^{-4} \xrightarrow{\text{CR}}$$

$$F'(t) = -4(2t+1)^{-5} \cdot \frac{d}{dt}(2t+1) = -4(2t+1)^{-5}(2) = -\frac{8}{(2t+1)^5}$$

$$47. y = \cos \sqrt{\sin(\tan \pi x)} = \cos(\sin(\tan \pi x))^{1/2} \Rightarrow$$

$$\begin{aligned} y' &= -\sin(\sin(\tan \pi x))^{1/2} \cdot \frac{d}{dx} (\sin(\tan \pi x))^{1/2} = -\sin(\sin(\tan \pi x))^{1/2} \cdot \frac{1}{2} (\sin(\tan \pi x))^{-1/2} \cdot \frac{d}{dx} (\sin(\tan \pi x)) \\ &= \frac{-\sin \sqrt{\sin(\tan \pi x)}}{2 \sqrt{\sin(\tan \pi x)}} \cdot \cos(\tan \pi x) \cdot \frac{d}{dx} \tan \pi x = \frac{-\sin \sqrt{\sin(\tan \pi x)}}{2 \sqrt{\sin(\tan \pi x)}} \cdot \cos(\tan \pi x) \cdot \sec^2(\pi x) \cdot \pi \\ &= \frac{-\pi \cos(\tan \pi x) \sec^2(\pi x) \sin \sqrt{\sin(\tan \pi x)}}{2 \sqrt{\sin(\tan \pi x)}} \end{aligned}$$

9. (6 pts) Find the linear approximation of the function $f(x) = \sqrt{1-x}$ at $a = 0$ and use it to approximate the numbers of $\sqrt{0.9}$ and $\sqrt{0.99}$.

Answer:

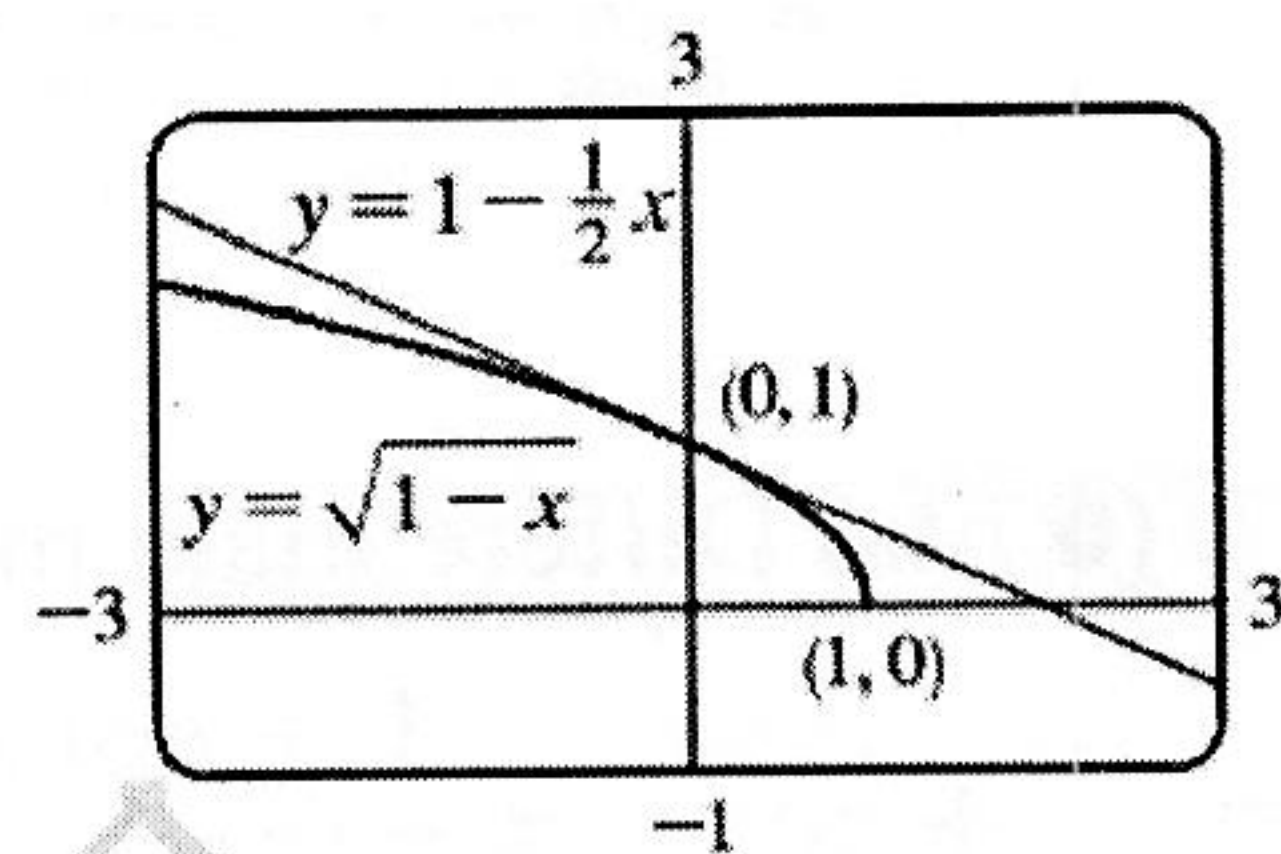
$$f(x) = \sqrt{1-x} \Rightarrow f'(x) = \frac{-1}{2\sqrt{1-x}}, \text{ so } f(0) = 1 \text{ and } f'(0) = -\frac{1}{2}.$$

Therefore,

$$\sqrt{1-x} = f(x) \approx f(0) + f'(0)(x-0) = 1 + \left(-\frac{1}{2}\right)(x-0) = 1 - \frac{1}{2}x.$$

$$\text{So } \sqrt{0.9} = \sqrt{1-0.1} \approx 1 - \frac{1}{2}(0.1) = \underline{0.95}$$

$$\text{and } \sqrt{0.99} = \sqrt{1-0.01} \approx 1 - \frac{1}{2}(0.01) = \underline{0.995}.$$



10. (6 pts) Find the absolute maximum and absolute minimum values of f on the given interval.

$$f(t) = 2 \cos t + \sin 2t, [0, \frac{\pi}{2}]$$

Answer:

$$f(t) = 2 \cos t + \sin 2t, [0, \pi/2].$$

$$f'(t) = -2 \sin t + \cos 2t \cdot 2 = -2 \sin t + 2(1 - 2 \sin^2 t) = -2(2 \sin^2 t + \sin t - 1) = -2(2 \sin t - 1)(\sin t + 1).$$

$$f'(t) = 0 \Rightarrow \sin t = \frac{1}{2} \text{ or } \sin t = -1 \Rightarrow t = \frac{\pi}{6}. \quad f(0) = 2, f(\frac{\pi}{6}) = \sqrt{3} + \frac{1}{2} \sqrt{3} = \frac{3}{2} \sqrt{3} \approx 2.60, \text{ and } f(\frac{\pi}{2}) = 0. \quad * \text{錯一個扣!}$$

So $f(\frac{\pi}{6}) = \frac{3}{2} \sqrt{3}$ is the absolute maximum value and $f(\frac{\pi}{2}) = 0$ is the absolute minimum value.

11. (6 pts) Verify the function satisfies the hypotheses of the Mean value Theorem on the given interval. Then find all numbers c that satisfy the conclusion of the Mean Value Theorem.

$$f(x) = 2x^2 - 3x + 1, [0, 2]$$

Answer:

$$f(x) = 2x^2 - 3x + 1, [0, 2]. \quad f \text{ is continuous on } [0, 2] \text{ and differentiable on } (0, 2) \text{ since polynomials are continuous and}$$

$$\text{differentiable on } \mathbb{R}. \quad f'(c) = \frac{f(b) - f(a)}{b - a} \Leftrightarrow 4c - 3 = \frac{f(2) - f(0)}{2 - 0} = \frac{3 - 1}{2} = 1 \Leftrightarrow 4c = 4 \Leftrightarrow c = 1, \text{ which}$$

is in $(0, 2)$.

12. (6 pts) $f(x) = (x^2 - 4)^3$

(a) Find the intervals of increase or decrease.

(b) Find the local maximum and minimum values.

(c) Find the intervals of concavity and the inflection points.

Answer:

(a) $f(x) = (x^2 - 4)^3 \Rightarrow f'(x) = 3(x^2 - 4)^2(2x) = 6x(x^2 - 4)^2$. Since $(x^2 - 4)^2$ is nonnegative, the sign of $f'(x)$ is determined by the sign of $6x$. Thus, $f'(x) < 0 \Leftrightarrow x < 0$ [$x \neq -2$] and $f'(x) > 0 \Leftrightarrow x > 0$ [$x \neq 2$]. So f is increasing on $(0, 2)$ and $(2, \infty)$, and f is decreasing on $(-\infty, -2)$ and $(-2, 0)$. By Exercise 3.3.79, we can say that f is increasing on $(0, \infty)$ and decreasing on $(-\infty, 0)$.

(b) f changes from decreasing to increasing at $x = 0$. Thus, $f(0) = -64$ is a local minimum value. local maximum doesn't exist

(c) $f''(x) = 6x \cdot 2(x^2 - 4)(2x) + (x^2 - 4)^2 \cdot 6 = 6(x^2 - 4)[4x^2 + (x^2 - 4)]$
 $= 6(x^2 - 4)(5x^2 - 4) = 6(x + 2)(x - 2)(\sqrt{5}x + 2)(\sqrt{5}x - 2)$

Interval	$x + 2$	$\sqrt{5}x + 2$	$\sqrt{5}x - 2$	$x - 2$	$f''(x)$	f
$x < -2$	-	-	+	-	+	concave up on $(-\infty, -2)$
$-2 < x < -\frac{2}{\sqrt{5}}$	+	-	-	-	-	concave down on $(-2, -\frac{2}{\sqrt{5}})$
$-\frac{2}{\sqrt{5}} < x < \frac{2}{\sqrt{5}}$	+	+	-	-	+	concave up on $(-\frac{2}{\sqrt{5}}, \frac{2}{\sqrt{5}})$
$\frac{2}{\sqrt{5}} < x < 2$	+	+	+	-	-	concave down on $(\frac{2}{\sqrt{5}}, 2)$
$x > 2$	+	+	+	+	+	concave up on $(2, \infty)$

There are inflection points at $(-2, 0)$, $(-\frac{2}{\sqrt{5}}, -\frac{4096}{125})$, $(\frac{2}{\sqrt{5}}, -\frac{4096}{125})$, and $(2, 0)$.

13. (6 pts) Find the horizontal and vertical asymptotes of the curve.

$$y = \frac{2x^2 + x - 1}{x^2 + x - 2}$$

Answer:

$$\begin{aligned} \lim_{x \rightarrow \pm\infty} \frac{2x^2 + x - 1}{x^2 + x - 2} &= \lim_{x \rightarrow \pm\infty} \frac{\frac{2x^2 + x - 1}{x^2}}{\frac{x^2 + x - 2}{x^2}} = \lim_{x \rightarrow \pm\infty} \frac{2 + \frac{1}{x} - \frac{1}{x^2}}{1 + \frac{1}{x} - \frac{2}{x^2}} = \frac{\lim_{x \rightarrow \pm\infty} (2 + \frac{1}{x} - \frac{1}{x^2})}{\lim_{x \rightarrow \pm\infty} (1 + \frac{1}{x} - \frac{2}{x^2})} \\ &= \frac{\lim_{x \rightarrow \pm\infty} 2 + \lim_{x \rightarrow \pm\infty} \frac{1}{x} - \lim_{x \rightarrow \pm\infty} \frac{1}{x^2}}{\lim_{x \rightarrow \pm\infty} 1 + \lim_{x \rightarrow \pm\infty} \frac{1}{x} - 2 \lim_{x \rightarrow \pm\infty} \frac{1}{x^2}} = \frac{2 + 0 - 0}{1 + 0 - 2(0)} = 2, \text{ so } y = 2 \text{ is a horizontal asymptote.} \end{aligned}$$

$$\begin{aligned} y = f(x) &= \frac{2x^2 + x - 1}{x^2 + x - 2} = \frac{(2x - 1)(x + 1)}{(x + 2)(x - 1)}, \text{ so } \lim_{x \rightarrow -2^-} f(x) = \infty, \\ \lim_{x \rightarrow -2^+} f(x) &= -\infty, \lim_{x \rightarrow 1^-} f(x) = -\infty, \text{ and } \lim_{x \rightarrow 1^+} f(x) = \infty. \text{ Thus, } x = -2 \end{aligned}$$

and $x = 1$ are vertical asymptotes. The graph confirms our work.

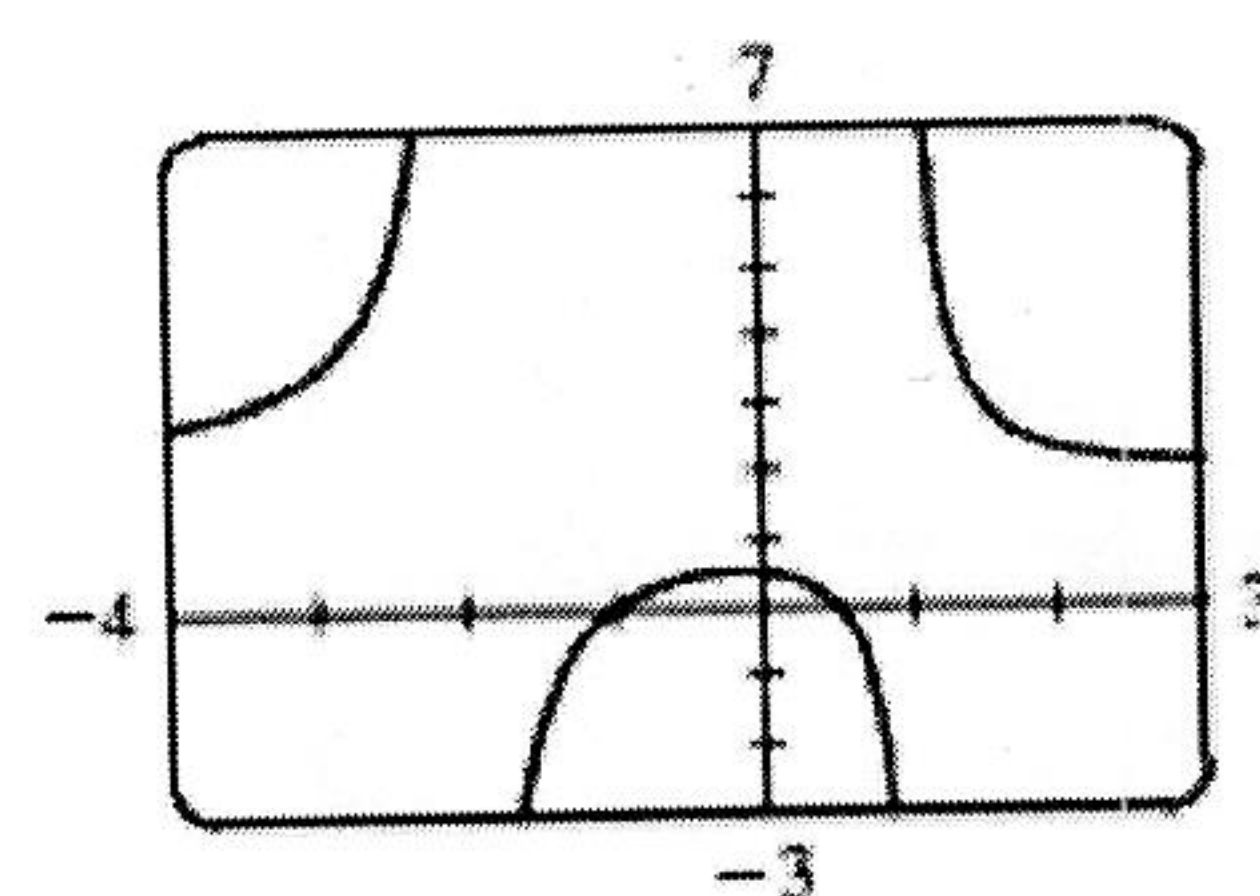
* 小地方寫錯只扣1 整體沒有扣2

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14. (6 pts) Find the limits at $x \rightarrow \infty$ and as $x \rightarrow -\infty$. Use the information, together with intercepts, to give a rough description of the graph.

$$y = (3 - x)(1 + x)^2(1 - x)^4$$

Answer:



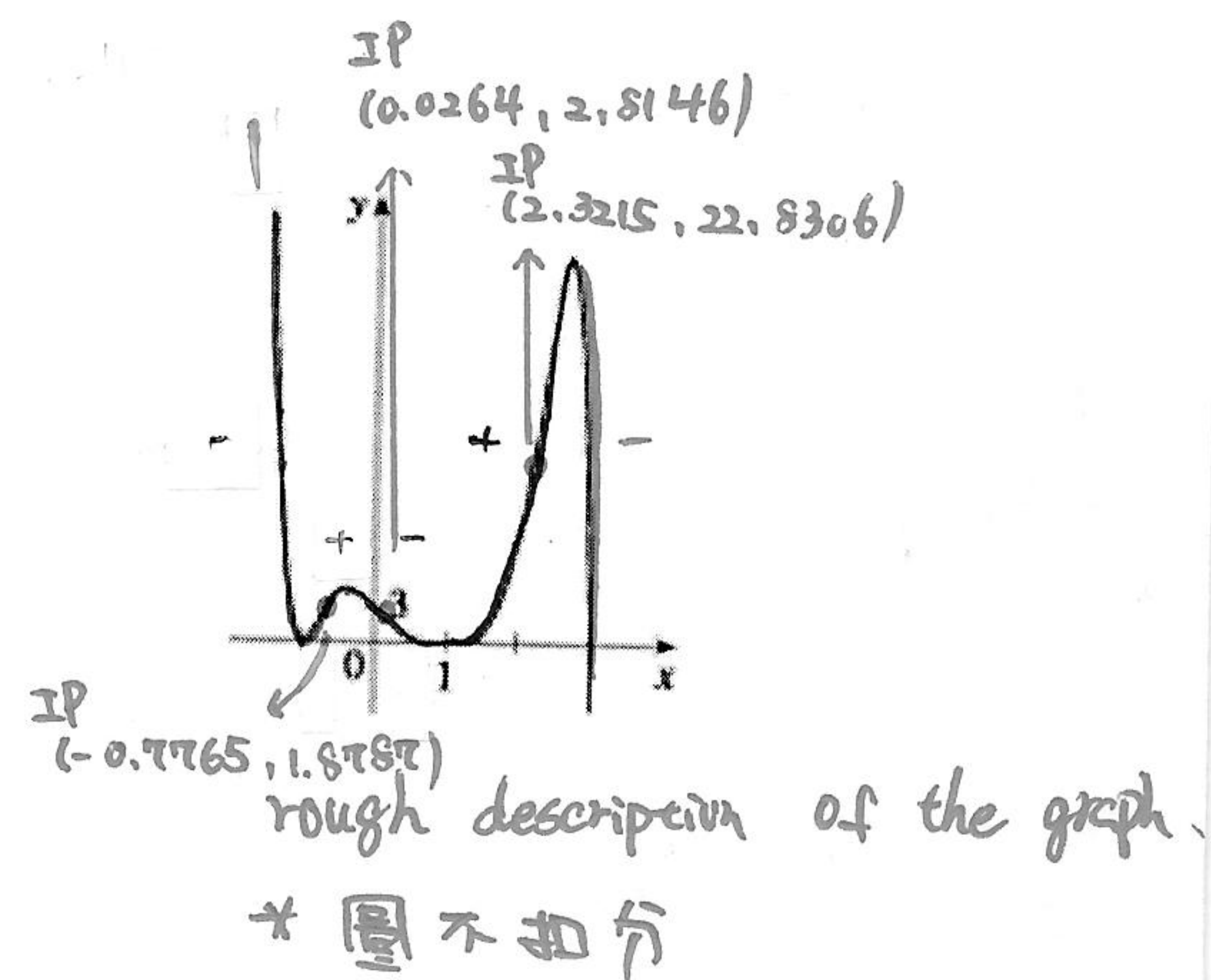
$y = f(x) = (3-x)(1+x)^2(1-x)^4$. The y-intercept is $f(0) = 3(1)^2(1)^4 = 3$.

The x-intercepts are 3, -1, and 1. There is a sign change at 3, but not at -1 and 1.

When x is large positive, $3-x$ is negative and the other factors are positive, so

$\lim_{x \rightarrow \infty} f(x) = -\infty$. When x is large negative, $3-x$ is positive, so

$\lim_{x \rightarrow -\infty} f(x) = \infty$.



$$f(x) = (3-x)(1+x)^2(1-x)^4$$

$$f'(x) = [-(1+x)^2 + (3-x)2(1+x)](1-x)^4 + (3-x)(1+x)^2 4(1-x)^3(-1)$$

$$= (1+x)(-3x+5)(1-x)^4 + 4(x-3)(1+x)^2(1-x)^3$$

$$= (1+x)(1-x)^3 [(-3x+5)(1-x) + 4(x-3)(1+x)]$$

$$= (1+x)(1-x)^3 (7x^2 - 16x - 7)$$

$$f''(x) = [(1-x)^3 + (1+x)3(1-x)^2(-1)](7x^2 - 16x - 7) + (1+x)(1-x)^3(14x - 16)$$

$$= (1-x)^2(-2-4x)(7x^2 - 16x - 7) + (1+x)(1-x)^3(14x - 16)$$

$$= (1-x)^2 [(-2-4x)(7x^2 - 16x - 7) + (1+x)(1-x)(14x - 16)]$$

$$= (1-x)^2 (-21x^3 + 66x^2 + 74x - 2)$$

$$= -2(1-x)^2 (21x^3 - 33x^2 - 37x + 1)$$

$$\frac{16 \pm \sqrt{16^2 - 4 \times 7 \times (-7)}}{14}$$

$$\frac{16 \pm \sqrt{452}}{14}$$

$$\frac{8 \pm \sqrt{113}}{7}$$

x	-1	-0.38	1	2.4
$f'(x)$	-	0	+	0
$f''(x)$			0	+
$f(x)$	0		0	0